Project: Worked examples
Title: Pipe pressure drop and pump duty
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

Friction and fitting losses through series pipe segments at a given flow, the static head, the system curve, the duty point against a pump curve entered as points, the affinity-law shift for a speed or impeller change, and the NPSH available at the pump suction.

$v = Q/(\pi D^2/4)$; $Re = \rho v D/\mu$; $f = 64/Re$ ($Re < 2300$) else Colebrook–White $1/\sqrt{f} = -2\log_{10}(\varepsilon/3.7D + 2.51/(Re\sqrt{f}))$ by fixed-point iteration (≤ 100 steps) with the Swamee–Jain explicit value as start and cross-check; $h_f = f\,(L + L_{eq})/D \cdot v^2/2g + K\,v^2/2g$; $H_{sys}(Q) = \Sigma dz + h_{out} + \Sigma h_f(Q)$ with $h_f \propto Q^2$ scaled from the design flow; duty point where $H_{pump}(Q) = H_{sys}(Q)$ (bisection); affinity: $Q \propto n\,D^3$, $H \propto n^2 D^2$; $NPSH_a = p_{atm}/\rho g + z_s - h_{f,suction} - p_v/\rho g$.

Ref: Crane Technical Paper 410 — Flow of fluids through valves, fittings and pipe
Ref: Çengel & Cimbala, Fluid Mechanics Ch. 8 — Internal flow: friction factor, minor losses, pump duty
Ref: CIBSE Guide C Section 4 — Flow of fluids in pipes and ducts (principles)
Ref: Kell, J. Chem. Eng. Data 20(1), 1975 — Density of liquid water

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
fluid	Fluid	custom (□ and □ given)		spec: Oil datasheet ISO VG 680 at 20 °C	any
T	Fluid temperature	20	degC	assumption: Ambient 20 °C.	0 ... 100 degC
rho_custom	Density (custom fluid)	888	kg/m^3	spec: Oil datasheet ISO VG 680	500 ... 2000 kg/m^3
mu_custom	Dynamic viscosity (custom fluid)	800	mPa*s	spec: Oil datasheet ISO VG 680 at 20 °C	0.1 ... 1000 mPa*s
Q	Design flow rate	1	m^3/h	spec: Transfer pump duty	0.01 ... 100000 m^3/h
segments	Pipe segments in series	1 row(s)		drawing: DRG-L-020 rev A	any
h_out	Pressure head required at the outlet	0	m	assumption: Open discharge.	0 ... 1000 m
v_max	Velocity limit (warning)	3	m/s	assumption: Practice limit.	0.5 ... 10 m/s
n_ratio	Speed ratio n_2/n_1 (affinity)	1		default	0.2 ... 2
D_ratio	Impeller diameter ratio D_2/D_1 (affinity)	1		default	0.7 ... 1.1
eta_pump	Pump efficiency at duty (for shaft power)	0.7		default	0.2 ... 0.95
p_atm	Atmospheric pressure (NPSH)	101.325	kPa	default	50 ... 110 kPa

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
z_s	Static head at the pump suction (+ flooded, − lift)	0	m	default	-10 … 100 m

3. Assumptions

A1. Steady, incompressible, fully developed flow; all segments carry the full design flow in series. (Darcy–Weisbach.)
A2. Fitting losses scale with velocity head (K method) or add pipe length (equivalent-length method); both may be used on one segment. (Crane TP-410 practice.)
A3. The system curve scales friction losses with Q² from the design flow (turbulent assumption); the static part is independent of flow. (H_sys = H_static + k Q².)
A4. Water properties are from published correlations at atmospheric pressure (0–100 °C); any other fluid is entered with its own density and viscosity and their provenance. (Kell 1975; Vogel fit; Hyland & Wexler 1983.)
A5. T (Fluid temperature) is taken as 20 degC. (Ambient 20 °C.)
A6. h_out (Pressure head required at the outlet) is taken as 0 m. (Open discharge.)
A7. v_max (Velocity limit (warning)) is taken as 3 m/s. (Practice limit.)

4. Working

[5] rho = rho_custom
= 888 kg/m³
= 888 kg/m³
Density (as entered)
[6] mu = mu_custom
= 800 mPa·s
= 800 mPa·s
Dynamic viscosity (as entered)
[10] v_s1 = transfer line: Q / (π D²/4)
= 1 m³/h / (π × 50² mm² / 4)
= 0.14147 m/s
Velocity, transfer line
[11] Re_s1 = transfer line: ρ v D / μ
= 888 × 0.14147 × 0.05 / 0.8
= 7.8516
Reynolds number, transfer line
[12] f_s1 = transfer line: Darcy friction factor
= 64 / Re (laminar)
= 8.1512
Friction factor, transfer line
[13] h_f_s1 = transfer line: f (L + L_eq)/D · v²/2g + K v²/2g
= 8.1512 × (40 + 0) / 0.05 × 0.0010204 + 0 × 0.0010204
= 6.6542 m
Head loss, transfer line
[14] h_f_total = Σ h_f
= 1 segment(s)
= 6.6542 m
Total friction and fitting loss at the design flow
[15] H_static = Σ dz + h_out
= 0 + 0
= 0 m
Static head

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[16] H_sys = H_static + h_f_total
= 0 m + 6.6542 m
= 6.6542 m
```
System head at the design flow
```
[17] dp_total = rho * 9.80665 m/s^2 * h_f_total
= 888 kg/m³ × 9.80665 m/s^2 × 6.6542 m
= 57.947 kPa
```
Friction pressure drop at the design flow
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[18] dp_sys = rho * 9.80665 m/s^2 * H_sys
= 888 kg/m³ × 9.80665 m/s^2 × 6.6542 m
= 57.947 kPa
```
System pressure at the design flow
```
[19] P_hyd = rho * 9.80665 m/s^2 * Q * H_sys
= 888 kg/m³ × 9.80665 m/s^2 × 1 m³/h × 6.6542 m
= 0.016096 kW
```
Hydraulic power at the design point
```
[22] v_peak = highest segment velocity
= 0.14147 m/s
= 0.14147 m/s
```
Highest velocity
```
[23] u_v = v_peak / v_max
= 0.14147 / 3
= 0.047157
```
Velocity utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
v_s1	0.14147 m/s	Velocity, transfer line
Re_s1	7.8516	Re, transfer line
f_s1	8.1512	f, transfer line
h_f_s1	6.6542 m	h_f, transfer line
h_f_total	6.6542 m	Σ h_f
H_static	0 m	Static head
H_sys	6.6542 m	System head at Q
dp_total	57.947 kPa	Friction Δp
dp_sys	57.947 kPa	System Δp
P_hyd	0.016096 kW	Hydraulic power
v_peak	0.14147 m/s	Highest velocity

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Velocity ≤ limit	0.047157	1	0.0472	

7. Applicability

WARNING — laminar: segments: transfer line is laminar (Re = 7.8516); f = 64/Re.

8. Verification

Template bs.pipe-pump@1.0.0 Content hash 2b0ac6ee757adf02 Engine calc-engine/1.0.0

Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0